

SDR-Based Emulation of Machine Learning-Enabled Spectrum Sharing in 5G Private Networks

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Abstract—As 5G private networks become increasingly prevalent, efficient spectrum-sharing strategies for collocated deployments are critical. This study evaluates supervised, unsupervised and semi-supervised machine learning (ML) approaches for interference prediction and dynamic frequency/channel assignment in collocated 5G networks operating in the 3.5 GHz band. Utilizing GNU Radio OFDM transceivers and Software Defined Radio (SDR)-based emulation on the COSMOS testbed, we develop a spectrum management framework that leverages ML for real-time spectrum monitoring and intelligent resource allocation. The system, powered by an XGBoost model with 85% accuracy, accurately predicts interference and adjusts network frequencies to enhance performance. Semi-supervised models also perform well, achieving similar results with 60% unlabeled data. Key performance indicators such as Packet Error Rate (PER) and Signal-to-Interference-plus-Noise Ratio (SINR) show significant improvements following frequency reassignment, validating the effectiveness of the model-driven approach. Designed for scalability, this system is adaptable to more complex environments involving multiple networks and channels.

Index Terms—Machine learning, GNU Radio, Spectrum Sensing, Spectrum Sharing

I. INTRODUCTION

The increasing demand for wireless connectivity [1], fueled by connected devices and the rise of autonomous industries, has exposed the shortcomings of traditional spectrum management methods. With the expansion of 5G private networks [2], efficient spectrum sharing has become essential, particularly in collocated deployments.

In spectrum sharing frameworks like the Spectrum Access System (SAS) in Citizens Broadband Radio Service (CBRS) band [3], SAS faces challenges like complexity and service interruptions in congested areas. Machine learning (ML) offers a solution by improving spectrum efficiency, reducing latency, and enabling real-time adjustments.

Software-defined Radios (SDRs) enable flexible prototyping for 5G networks, while Orthogonal Frequency Division Multiplexing (OFDM) supports the high data rates crucial for 5G performance. The NTIA promotes SDR advancements in spectrum sharing [4], and recent 3GPP standards encourage embedding ML in Radio Access Networks (RANs) to enhance spectrum management.

This paper focuses on three primary objectives: (i) Developing a multi-class ML model to accurately predict interference

levels by analyzing various network parameters; (ii) Creating a spectrum manager framework and dynamic spectrum allocation strategy that minimizes interference and maximizes network performance; and (iii) Validating the proposed system model via simulation and real-world experiments in the 3.5 GHz band on the COSMOS testbed [5] to ensure its practical applicability.

The paper proceeds as follows: Section II details the system model, Section III reviews relevant literature, Section IV describes the experimental setup, Section V discusses the results, and Section VI concludes with the key contributions.

II. SYSTEM MODEL

The proposed system model evaluates and optimizes spectrum sharing between two co-located 5G private networks using a machine learning (ML) approach to predict interference and dynamically adjust frequency allocations. The implementation is based on an Orthogonal Frequency-Division Multiplexing (OFDM) setup, with applications to both the 5G NR n78 band [6] and the CBRS band.

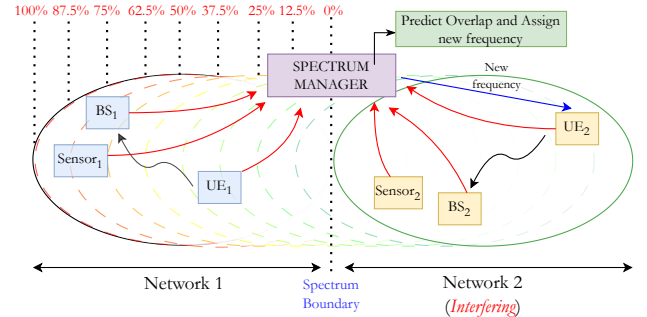


Fig. 1. Spectrum Manager Framework.

As shown in Figure 1, the model consists of two base stations (BS1 and BS2) and two users (UE1 and UE2) within separate but spatially collocated 5G networks. The base stations and UEs communicate using OFDM, a technology essential for achieving high data rates and managing interference. Sensors co-located with the base stations monitor spectrum use and detect interference, providing critical data for the Spectrum Manager.

The networks, Network 1 and Network 2, may interfere if their frequencies overlap. The percentages in Figure 1 indicate the degree of frequency overlap, highlighting the spectral relationship between the networks rather than a spatial one. This study focuses solely on downlink scenarios.

The Spectrum Manager uses an ML model to predict interference and manage frequency allocations. UEs transmit OFDM packets to base stations, where SINR and PSD are estimated. Sensors monitor channelization and detect interfering frequencies. This data is aggregated and analyzed by the Spectrum Manager, which predicts interference and assigns new frequencies as needed.

The interference overlap percentage is predicted using the following formula:

$$I_{\text{pred}} = f(\text{NW1 metrics}, \text{NW2 metrics})$$

Where:

- I_{pred} is the predicted percentage of overlap and $f(\cdot)$ is the function learned by the ML model
- NW1 metrics includes the packet error rate, SINR, 64 FFT bins, and frequency observations from 10 sensor channels for Network 1.
- NW2 metrics includes the corresponding metrics for Network 2.

III. RELATED WORK

Machine learning (ML) plays a crucial role in optimizing Radio Resource Management (RRM) for 5G networks, addressing issues like throughput prediction, network coverage, and spectral efficiency. However, many existing approaches struggle with real-time adaptability, a critical factor for private 5G networks. While reinforcement learning and deep neural networks have been explored for spectrum access and power control, their complexity often hinders real-time performance.

ML-based solutions for dynamic spectrum allocation and interference mitigation in private 5G networks show promise but face challenges with real-time implementation, as highlighted by [7] and [8]. Issues like user congestion and TDD synchronization in CBRS deployments further complicate spectrum management [9]. Research on cognitive radio networks and systems like OpenSAS has demonstrated the potential of ML for spectrum monitoring, but real-world applicability remains limited due to high computational demands and reliance on simulations [10], [11]. Our work addresses these gaps by employing simpler, real-time ML models for spectrum management. Further, earlier works have been predominantly simulation-based studies, while this paper tackles the problem via SDR-based emulation.

Authors of [12] introduced an ML-based spectrum sharing technique focused on optimizing Contention Window (CW) settings in industrial settings but lack real-world validation. Moreover, the authors only investigate supervised ML models. In addition, we explore semi-supervised and unsupervised models to address the reliance on labeled data, offering simpler models more suited for real-time applications.

ML-driven spectrum sharing faces hurdles like data quality, computational overhead, and real-time decision-making requirements. Complex ML models may not meet the demands of dynamic environments like CBRS, where effective spectrum sensing and interference detection are critical, especially in sensitive regulatory contexts.

IV. METHODOLOGY

A. Experiment Setup

Experiments were conducted on the COSMOS testbed using Software Defined Radios (SDRs) to implement 5G spectrum sharing strategies within the 3.5 GHz band. USRP B210 SDRs were used for OFDM transmission and reception, serving as base stations (BS1 and BS2) and user equipment (UE1 and UE2) across two networks. N210 SDRs [13] functioned as sensors co-located with the base stations, monitoring spectrum usage and interference. GNU Radio version 3.8 [14] facilitated signal processing, with experiments conducted on Linux Ubuntu 20.04 for stability.

Ten distinct network topologies were developed to capture a range of operational scenarios, including spatial variations and potential SDR malfunctions, to simulate real-world conditions, with Network 2 designated as the interfering network. The experiments were automated using shell scripts that managed network nodes, controlled transmissions, receptions, and spectrum sensing tasks, and collected key metrics such as PER, SINR, and spectral PSD across overlap scenarios from 0% to 100% in 12.5% increments. These scenarios resulted in nine distinct classes for machine learning training, representing the varying overlap conditions.

B. OFDM Transceiver Implementation

Our system uses Orthogonal Frequency Division Multiplexing (OFDM) for data transmission, a key technology for managing high-capacity and flexible communication in 5G networks. OFDM's ability to handle frequency-selective fading and narrowband interference makes it suitable for the dynamic spectrum sharing scenarios addressed in this study.

We implemented a custom OFDM transceiver using GNU Radio, which allowed us to adjust key parameters. The transmitter converts data into OFDM frames with BPSK modulation, while the receiver handles synchronization, frequency correction, and channel estimation to ensure reliable data decoding. Different 'sync words' were used for each network (NW1 and NW2) to maintain network isolation, critical in our experiments with private 5G networks.

C. Spectrum Sensing

The sensors monitored a 10 MHz bandwidth using energy detection, with a fixed threshold of -70 dB. This setup divided the frequency range [3.495 GHz, 3.504 GHz] into ten 1 MHz channels, allowing assessment of how different spectrum portions are affected by various overlap scenarios.

Traditional SINR calculation faced limitations, particularly in scenarios with high overlap between networks. The challenge was in differentiating between desired signals and

interference, which could lead to inaccurate readings. To address this, an updated SINR formula, referred to as Effective SINR (eSINR) or Interference-Aware SINR (IA-SINR), was introduced. This formula accounts for the significant presence of interference from overlapping signals:

$$\text{eSINR} = \frac{(1 - \alpha) \cdot P_{\text{desired}}}{P_{\text{interference}} + P_{\text{noise}}}$$

where P_{desired} is the power of the desired signal within the bandwidth of interest, $P_{\text{interference}}$ is the power of the interfering signals within the bandwidth of interest, P_{noise} is the power of the noise within the bandwidth of interest, and α is a coefficient representing the degree of overlap, with values closer to 1 indicating higher overlap.

This eSINR formula adjusts the interference impact based on the overlap percentage. This approach provides a more accurate reflection of signal quality by considering the extent of overlap directly in the interference calculation. Future work could explore adaptive thresholding or ML-based methods for improved signal detection.

D. Spectrum Manager

The Spectrum Manager dynamically manages frequency assignments and reallocations during network operations using an ML model to predict interference and adjust frequencies in real-time.

Upon detecting interference, the Spectrum Manager shifts the center frequency of the interfering network (Network 2) by 125 kHz, iteratively repeating this process until the interference is minimized and the packet error rate (PER) meets quality-of-service (QoS) criteria. This 125 kHz shift aligns with the 9-class problem structure, where each class corresponds to a specific frequency band. This systematic approach is similar to the Spectrum Access System (SAS) in the CBRS band [15], but with a more deterministic frequency adjustment strategy, which ensures reliable QoS but may lead to slower adaptation.

The frequency reassignment strategy adjusts the center frequency f_{nw2} of Network 2 to reduce the predicted interference I_{pred} . The adjustment follows this rule:

$$f_{\text{nw2}}(n+1) = f_{\text{nw2}}(n) + \Delta f$$

where $\Delta f = 125$ kHz is the adjustment step size. This process continues until I_{pred} is minimized to meet the required QoS, such as a satisfactory packet error rate.

E. Data Preparation and Machine Learning Strategy

Our machine learning (ML) model development involved thorough data collection, preprocessing, and experimentation to ensure robustness and accuracy.

Missing data was imputed using K-Nearest Neighbors (KNN) to preserve the data's structure and minimize bias. Datasets were created with 2, 4, and 9 class distributions, representing overlap percentages from 0% to 100%. Principal Component Analysis (PCA) reduced dimensionality while retaining over 95% of the variance.

Exploratory Data Analysis (EDA) and preprocessing were done using Python and scikit-learn. The model development began with binary classification (0% and 100% overlap), expanding to 4-class and 9-class problems for effective multi-class training. Key metrics like PER, SINR, and spectral PSD were collected using automated shell scripts controlling network nodes.

We tested various supervised learning models [16], including Logistic Regression, KNN, Decision Trees, Random Forest, and Naïve Bayes, focusing on multi-class classification. Unsupervised methods [16] like K-Means and Hierarchical Clustering were also explored but proved less effective for distinguishing overlap levels. To enhance performance, semi-supervised techniques [17] such as Label Propagation and Label Spreading were incorporated, bridging the gap between supervised and unsupervised learning, especially in cases with limited labeled data.

V. RESULTS

A. Exploratory Data Analysis

The analysis examined the impact of frequency overlap and spatial variations using ten distinct network topologies, on key metrics such as Packet Error Rate (PER), Signal-to-Interference-plus-Noise Ratio (SINR), Received Signal Strength Indicator (RSSI), Average Received Power, and Fast Fourier Transform (FFT) bins. Moreover, eleven new features were engineered, including power ratios and spectral entropy, to enhance the model's focus on critical signal characteristics like spectral energy distribution and SINR metrics. This feature engineering process highlighted correlations in FFT bins and signal power, significantly improving the model's ability to predict overlap conditions and optimize spectrum management. These insights were crucial for informing the design and optimization of the machine learning models used in this study.

a) *Packet Error Rate (PER)*: In all datasets, increasing spectral overlap between NW1 and NW2 consistently led to higher Packet Error Rate (PER), highlighting the negative impact of interference on data transmission. Figure 2 shows PER rising significantly from 0% to 100% overlap, with near-maximum levels in full overlap scenarios. In the 4-class dataset, PER increased with overlap, showing less variability and more consistent packet loss. Similarly, the 9-class dataset shows a steady PER increase, stabilizing around 0.8 to 1 as overlap approached 100%, indicating more frequent and stable packet errors with greater overlap.

b) *Sensor SINR*: The analysis of Sensor SINR revealed an unexpected trend where SINR values increased with higher overlap, which contradicts the expectation that SINR should decrease as interference rises. This anomaly, observed across all datasets, stems from limitations in the traditional SINR calculation using GNU Radio, where interference is inaccurately treated as part of the desired signal, leading to misleadingly high SINR values even at 100% overlap. To address this, the Effective SINR (eSINR) formula was proposed, adjusting

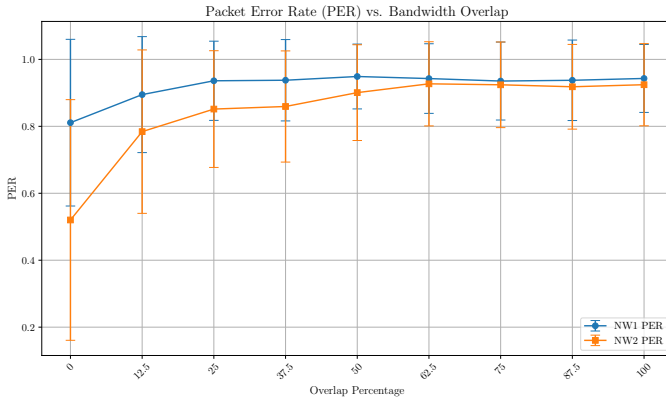


Fig. 2. Packet Error Rate versus Bandwidth Overlap

for the degree of overlap and providing a more accurate representation of signal quality in high-interference scenarios.

c) *Received Signal Strength Indicator (RSSI)*: RSSI values showed a slight decline as overlap increased, indicating that interference had some impact, though less pronounced compared to other metrics. In the 2-class dataset, there was a noticeable decrease in RSSI at 100% overlap, highlighting the challenge of distinguishing between desired signals and interference. The 4-class and 9-class datasets demonstrated consistent decreases in RSSI for both NW1 and NW2 as overlap increased, aligning with the expectation that higher interference degrades signal quality.

d) *FFT Bins Analysis*: As overlap increased, the median power values in the FFT bins generally rose, especially in the 100% overlap scenario, indicating a clear impact of interference. This trend was accompanied by greater variability and more outliers, highlighting the complex nature of signal interference. In the 2-class dataset, the increase in median power levels at 100% overlap was subtle. However, in the 4-class and 9-class datasets, the rise in median power became more pronounced, particularly in the higher FFT bins, with a broader interquartile range (IQR) and more outliers in higher overlap conditions, illustrating the varying impact of overlap across different bins.

e) *Channel-Specific Observations*: Channels near the central frequency (3.499 GHz to 3.500 GHz) were more sensitive to overlap, showing higher signal detection and pronounced interference. Conversely, edge channels (e.g., 3.495 GHz and 3.504 GHz) remained stable with fewer fluctuations, indicating lower susceptibility to interference. The presence of outliers in some channels, especially under lower overlap, suggests sporadic high-frequency events likely due to noise or specific interference. These results align with our expectations, given that Network 1's center frequency was set at 3.5 GHz, with experiments focused on overlaps near this frequency.

B. ML Model Performance Insights

The ML models were trained on a KNN-imputed dataset containing 160 features. Also, to further enhance the dataset, we performed feature engineering, expanding the feature set

to 171, and applied feature selection using loading scores to refine the feature set; PCA was implemented too, retaining 95% of the variance, which reduced the feature set to 34. We maintained 1000 observation points per class across all datasets, deliberately avoiding imbalanced dataset challenges to prevent biased results.

1) *Supervised Model Approach*: We trained a range of machine learning models, from simple algorithms such as logistic regression and random forest, to more complex ensemble models and advanced boosting models like XGBoost and CatBoost. The top-performing models were selected based on accuracy and F1 score as the primary evaluation metrics.

In both the 2-class and 4-class datasets, the top-performing models consistently demonstrated high accuracy and robustness. For the 2-class dataset, the CatBoost Classifier led with a test accuracy and F1 score of 97.21%, followed closely by Random Forest (96.00%) and Logistic Regression (95.50%), particularly when enhanced with feature engineering. In the 4-class dataset, the Light Gradient Boosting Machine (LightGBM) achieved the highest accuracy at 94.57%, followed by Random Forest at 92.13% and Extra Trees Classifier at 92.31%, both benefiting from KNN imputation and feature engineering.

For the more complex 9-class dataset, the Extreme Gradient Boosting (XGBoost) model demonstrated the highest performance, achieving a test accuracy and F1 score of approximately 85.37%. Random Forest followed with a test accuracy of 78.88%. Logistic Regression showed relatively strong performance with a test accuracy of 77.61%.

As the number of classes increased, the models' complexity grew, leading to reduced classification accuracy, particularly in mid-range overlap scenarios such as 25%, 37.5%, 50%, 62.5%, and 75%. However, the models were able to clearly distinguish between extreme cases like 0% and 100% overlap, highlighting the challenges of classifying more granular overlap conditions based on the collected metrics.

Insights on XGBoost: The XGBoost model, renowned for its strong performance in complex classification tasks, proved highly effective in the 9-class overlap dataset—the most challenging in our study. Its gradient boosting framework, which iteratively improves weak learners, allowed XGBoost to accurately classify different overlap conditions by capturing complex interactions between key features such as 'S2Channel8', 'S2Channel7', and 'NW2sensorSINR'. These features, related to specific channel observations and SINR measurements, were crucial in driving the model's performance. While the model successfully differentiated most classes, subtle variations between certain neighboring classes—like Class 5 (62.5% overlap), Class 6 (75% overlap), and Class 7 (82.5% overlap)—led to some misclassifications, as suggested by the confusion matrix. This indicates that refining feature selection or increasing data diversity could further enhance the model's accuracy.

Additionally, the model's robustness in handling high-dimensional data was validated by its strong micro-average and macro-average ROC curves, with AUC values of 0.99

and 0.98, respectively. To assess generalization, XGBoost was tested on unseen data with granular overlap cases (e.g., 30%, 10%) and achieved a “pseudo accuracy” of 81.28%, demonstrating its ability to capture general trends despite the complexity of precise predictions.

2) *Unsupervised Model Approach*: In addition to the supervised approaches, we explored unsupervised learning methods, motivated by the high cost of labeling data and the need for real-time implementation, where an unsupervised model could offer more flexibility. Given the nature of our dataset, we aimed to investigate whether unsupervised clustering techniques, such as K-Means, could naturally group data points into clusters that aligned with the expected interference overlap classes. This was based on the assumption that overlap conditions would lead to distinct clusters.

However, K-Means clustering struggled to effectively separate the classes, as evidenced by low silhouette scores, indicating poor cluster cohesion and significant overlap between the clusters. Despite some differentiation in overlap conditions, the clusters formed were not tightly defined, with assigned labels appearing somewhat random. Similarly, DBSCAN presented challenges due to extensive hyperparameter tuning requirements, making it less practical for real-time applications.

Agglomerative clustering showed more promise when applied to a simplified 2-class dataset, yielding moderate results. The silhouette score of 0.2508 and a Davies-Bouldin Index of 1.6567 suggested limited cluster coherence, though some degree of class distinction was observed. These results indicate that, while unsupervised models can offer some insights, clustering alone may not suffice in highly overlapping datasets. Investigating semi-supervised learning methods could enhance performance by leveraging both labeled and unlabeled data, providing a more nuanced understanding of the overlap conditions.

3) *Semi-supervised Model Approach*: We explored three semi-supervised learning methods from Python’s scikit-learn library: Self-Training, Label Propagation, and Label Spreading. Self-Training expanded the labeled dataset by iteratively adding pseudo-labels to the training set, while Label Propagation and Spreading used graph-based techniques to infer labels for unlabeled data points. Label Spreading proved more robust in handling noise due to its normalized edge weights, making it more suitable for complex datasets.

Our experiments yielded impressive accuracies despite feeding in 60% of unlabeled datapoints: 96.35% for 2-class, 89.85% for 4-class, and 84.37% for the 9-class dataset. However, these methods faced limitations in real-time applications, as they require retraining when new data arrives, making them less practical for low-latency environments like wireless networks. The distinction between inductive (generalizable to unseen data) and transductive (focused on observed data) SSL models also highlighted challenges in generalization for real-time scenarios.

Semi-supervised models require careful tuning, such as adjusting the regularization factor in Label Spreading. Moreover, challenges in representative data selection and retraining

complexities motivate future research into deeper neural networks and reinforcement learning, which offer online training capabilities essential for real-time spectrum sharing in 5G networks.

4) *Real-time Performance of Model*: The real-time performance of our ML models was evaluated on the COSMOS testbed, focusing on their impact on Packet Error Rate (PER) and Effective SINR (eSINR). The initial tests involved setting the interfering network’s center frequency to 3.5 GHz to simulate a worst-case overlap scenario. As shown in Table I, both supervised and semi-supervised models were tested, with results indicating strong performance for models like XGBoost, LightGBM, and Self-Training Classifier.

Test accuracy was calculated using offline datasets with train/test splits, while real-time accuracy was assessed in live scenarios with varying packet transmission counts (2k, 5k, 10k, 15k). In the first phase, using real-time data with granular overlaps (e.g., 30%, 40%, 60%) unseen by the model at training, XGBoost achieved a pseudo accuracy of 70.69%, demonstrating strong generalization. In the second phase, applying a frequency reassignment strategy to reduce spectral overlap to 0%, XGBoost maintained an effective accuracy of 79.17%, despite some discrepancies likely caused by adjacent channel interference. Overall, XGBoost handled dynamic conditions well, proving its potential for real-time spectrum sharing.

TABLE I
PERFORMANCE OF ML MODELS

Model Name	Test-set Accuracy	Real-time Accuracy
Extreme GB	0.8537	0.7917
Light GBM	0.8533	0.7250
Self Training	0.8512	0.7357
Label Propagation	0.8142	0.6214
Label Spreading	0.8127	0.5814
Random Forest	0.7872	0.6013

For semi-supervised models, we initially unlabelled 60% of the dataset for training, which produced promising test accuracy. To further assess real-time performance, we decreased the unlabelled data to 40%, and despite this, both the Self-Training classifier and Label Propagation achieved competitive real-time accuracies, comparable to supervised models, highlighting their utility in ‘labelled data’-scarce environments. However, the need to retrain the model when new data arrives poses a computational challenge in latency-sensitive settings like private 5G networks.

XGBoost and semi-supervised techniques delivered strong accuracy with reduced computational overhead, essential for real-time spectrum management. LightGBM, while slightly less accurate (72.50% in real-time), proved more efficient due to its histogram-based approach and lower memory usage, making it ideal for large-scale applications. While deep learning may excel in complex environments, the simplicity and

flexibility of XGBoost make it a strong candidate for smaller-scale, real-time deployments.

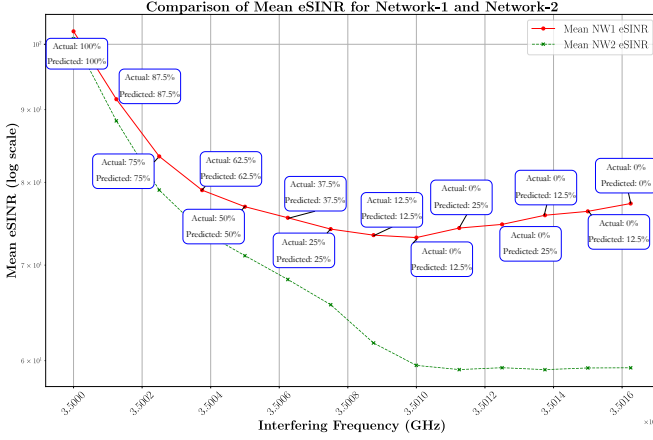


Fig. 3. Mean eSINR comparison & XGBoost predictions for Network 1 and Network 2 at different interfering frequencies.

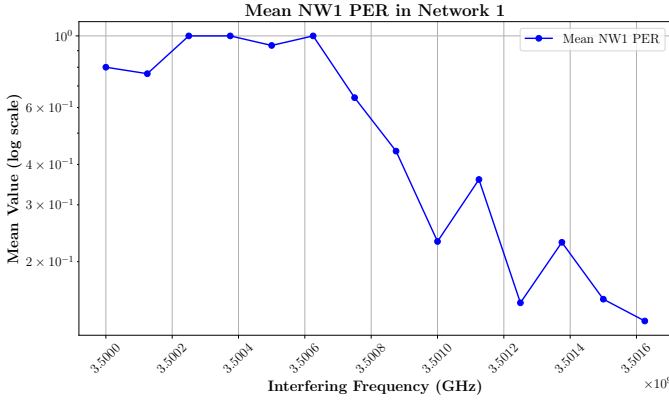


Fig. 4. Packet Error Rate in Network 1 at different interfering frequencies.

Figures 3 and 4 illustrate how mean eSINR and mean PER evolve as spectral overlap decreases. The mean eSINR and mean PER represent the average Effective SINR and Packet Error Rate, respectively, calculated across multiple observations for each interfering frequency. Figure 3 shows mean eSINR values for both networks across different interfering frequencies, with percentages indicating actual and predicted overlaps. As the interfering frequency shifts from Network 1's operating frequency, spectral overlap decreases, leading to improved eSINR, demonstrating reduced interference and better signal quality. Similarly, Figure 4 shows a steady decrease in mean PER for Network 1 as the interfering frequency moves away, indicating fewer packet errors and highlighting the model's effectiveness in managing interference and enhancing network performance.

VI. CONCLUSION

This study demonstrated the effectiveness of machine learning models, including XGBoost and semi-supervised meth-

ods like Label Spreading, for interference mitigation and spectrum sharing in 5G private networks using SDR-based emulation on the COSMOS testbed. XGBoost, with 85.37% accuracy, was instrumental in predicting interference and guiding frequency reassignment, leading to improvements in key metrics like Packet Error Rate (PER) and Effective SINR (eSINR). Although XGBoost showed the best performance, semi-supervised models also proved effective, even with 60% unlabeled data. The system's scalability suggests it could handle more complex scenarios, with potential future work focusing on increased interference, additional base stations, spectrum scheduling, and optimization using reinforcement learning.

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